Supavit Pitaknarongphorn

Prof. Gershon Weltman

ENGR 183EW

2 December 2024

Postmortem of Uber Fatal Crash, 2018

Abstract

Autonomous vehicle (AV) technology promises long-term reductions in traffic fatalities, accessibility to disabled people, and environmental benefits; however, the 2018 Uber fatal crash in Tempe, Arizona represents a pivotal case in the ethics and safety of AV technology. The incident occurred when a self-driving Uber, operating in autonomous mode with a safety driver, struck and killed a pedestrian as she crossed the street with her bicycle. This event exposed significant technical and ethical lapses, including a flawed algorithm, insufficient safety measures, employee negligence, and over-reliance on imperfect technology by Uber's management. From an ethical perspective, this case raises questions about the prioritization of profit over safety in the AV industry and the moral responsibilities of corporations when deploying unproven technologies in public settings. Applying the deontological framework, which emphasizes duty and adherence to ethical rules, the crash underscores Uber’s failure to ensure robust safety protocols and adequate human oversight. The intended audience of this postmortem is the National Highway Traffic Safety Administration (NHTSA) to implement stricter safety standards and increase regulatory oversight to ensure public safety. The Uber fatal crash serves as a reminder that human lives must remain the central consideration in engineering decisions, especially when introducing disruptive technologies into public spaces.

Background

The 2018 Uber fatal crash in Tempe, Arizona, marked a significant event in the development and public perception of AV technology. On the night of March 18th, 2018, a self-driving Uber Volvo XC90 struck and killed Elaine Herzberg, a pedestrian crossing a multi-lane road while pushing her bicycle. This incident, the first known fatality involving a fully autonomous vehicle, underscored the technological and ethical challenges associated with deploying AVs in public settings (National Transportation Safety Board [NTSB], 2019). Uber had been testing its self-driving cars in multiple cities, including Tempe, under a program designed to advance AV technology and gain a competitive edge in the ridesharing industry.

At the time of the crash, investigations revealed that the vehicle was operating in autonomous mode with a human safety driver, Rafaela Vasquez, behind the wheel. Initial investigations revealed that the vehicle's sensors detected Herzberg approximately six seconds before impact but failed to correctly classify her as a pedestrian or recognize her path as a collision risk (McCausland 2019). Detecting Herzberg as a fast moving object, the vehicle registered the need for an emergency stop just 1.3 seconds before the crash. However, that was not enough to avoid the moving object at a walking speed, resulting in a fatal collision.

Uber had disabled the emergency braking system during the test, relying instead on the safety driver to intervene if necessary. However, the system was also not designed to alert the driver when encountering dangers. In addition, the safety driver was distracted, reportedly streaming a television show on her phone, and did not take control in time to avoid the collision (Hawkins, 2018). The fatality revealed substantial flaws in Uber's AV system, including inadequacies in object detection algorithms, insufficient safety redundancies, and the absence of an effective mechanism to monitor driver attentiveness.

The incident also brought to light broader systemic issues in the AV industry. In an attempt to compete in the fast-growing industry, Uber prioritized rapid advancement over safety assurance. Uber had reportedly reduced its safety driver staffing from two to one in the months leading up to the crash and scaled back the vehicle's ability to perform emergency stops autonomously to prioritize smoother rides for passengers (NTSB, 2019). Internal pressures to accelerate the development of its AV program may have compromised safety protocols. Furthermore, the vehicle had a history of disengagements—situations where human drivers needed to take control—signaling that the technology was not yet mature enough for complex urban environments.

Regulatory oversight was another critical factor in the crash. Arizona, a hub for AV testing, had established lenient policies to attract companies like Uber. The state allowed AV testing with minimal reporting requirements and placed significant responsibility on corporations to self-regulate. This laissez-faire regulatory environment enabled Uber to conduct tests without demonstrating compliance with stringent safety standards, raising questions about the adequacy of public protections in the race to develop AV technologies (Neuman, 2018).

In the aftermath, Uber temporarily suspended its self-driving program and faced intense scrutiny from regulators, the public, and the media. The NTSB's final report attributed the crash to a combination of technological failures, inadequate safety protocols, and human error. Uber's organizational culture and prioritization of rapid development over rigorous safety measures were also criticized. Although the company implemented changes to its AV program, including improved driver training and enhanced vehicle sensors, the incident had far-reaching implications for the AV industry as a whole (NTSB, 2019). In the end, Uber made the decision to

end the AV technology research permanently. The Uber fatal crash catalyzed debates about the ethical and legal responsibilities of companies developing AVs.

The Engineering Failure

At the heart of the engineering failure was the vehicle's inability to reliably detect and classify objects. According to the National Transportation Safety Board (NTSB), Uber radio detection and ranging (RADAR) detected Herzberg approximately six seconds before impact but failed to identify her as a pedestrian pushing a bicycle. The system alternated between misclassifying her as an unknown object, a vehicle, and a bicycle with varying predicted paths (NTSB, 2019). This confusion in classification resulted in the vehicle failing to predict a collision accurately and initiate evasive maneuvers.

Another critical failure was the lack of an active emergency braking system while the vehicle was in autonomous mode. Uber had disabled the Volvo's factory-installed emergency braking feature to avoid conflicts with its proprietary system. Instead, the system relied entirely on the safety driver to intervene in emergencies (NTSB, 2019). However, as investigations revealed, this design decision created a significant safety gap. The vehicle's system registered the need for an emergency stop just 1.3 seconds before the crash, too late for the human safety driver, who was distracted, to respond effectively.

The engineering design also failed to account for the critical role of the safety driver. Uber's system lacked effective mechanisms to monitor the driver's attentiveness. The crash investigation revealed that the safety driver, Rafaela Vasquez, had been streaming a television show on her smartphone at the time of the incident (Hawkins, 2018). The absence of an alertness monitoring system, such as eye-tracking technology or similar safeguards, allowed for prolonged driver inattention during testing, significantly increasing the risk of human error.

The Uber system employed a parameter known as "action suppression," designed to reduce false alarms. This parameter delayed the system's response to detected obstacles to confirm a collision risk (NTSB, 2019). However, in practice, this calibration compromised safety by inhibiting timely responses to genuine threats. The system's inability to promptly classify Herzberg as a pedestrian and determine her path exacerbated the failure of action suppression, preventing the system from triggering necessary evasive actions.

Uber's testing and development protocols revealed a pattern of prioritizing rapid advancement over safety assurance. Internal documents suggested that Uber engineers were aware of recurring disengagements—situations requiring human intervention—but failed to address these adequately. Furthermore, the decision to reduce the number of safety drivers from two to one per vehicle diminished a critical layer of oversight, leaving no redundancy to mitigate human or system errors (Neuman, 2018). The testing environment itself also presented challenges, as the AV was deployed in complex urban settings where unpredictable pedestrian behavior is common.

The incident also underscored systemic gaps in regulatory compliance and safety oversight. Arizona, where the crash occurred, had minimal AV testing requirements, allowing companies to operate with significant autonomy (NTSB, 2019). Uber exploited this permissive regulatory environment to conduct tests without meeting stringent safety benchmarks. This regulatory leniency allowed engineering flaws to persist unaddressed until they culminated in a fatality.

Ethical Analysis

Using Deontological ethics framework, it is evident that Uber's decisions were unethical, resulthing in the fatal crash of 2018. Deotological framework consists of Universality and

Reciprocity Principles working in tandem. Universality Principle states that an action is ethical if it adheres to a maxim that acts like a universal law and should be followed no matter the time and place. Reciprocity Principle states that an action is ethical only if it is also acceptable when everyone else does it as well. Deontological framework fits particularly well in the AV industry since a new, fast-growing industry like the AV industry requires a universal standard that must be followed by everyone, and Deontological framework can provide that universal standard. Applying the Universality Principle, it is clear that the maxim every AV company should follow is that everyone must priritize public safety over profit. When Reciprocity Principle is also applied, it is obvious that it is wrong for AV companies to expedite development process and overlook safety features since there will be a large number of deaths and injuries if everyone acted in that manner.

The crash raises ethical dilemmas surrounding responsibility for the actions of autonomous systems. The vehicle's inability to correctly classify Elaine Herzberg as a pedestrian or predict her trajectory stems from flawed algorithmic design. However, the deontological framework mandates that companies must prioritize public safety, which in this case means that systems must be properly tested before deploying the technology and risking public safety. If every company deploys inadequately tested technology, the public will be in danger. Responsibility for these failures lies with Uber’s engineers and decision-makers, who allowed a system with known limitations to operate in public spaces.

Uber’s organizational culture also played a significant role in the ethical failures leading to the crash. A deontological perspective would critique Uber for failing to maintain ethical standards within its corporate practices. Internal reports suggest that the company prioritized rapid technological progress and market competition over thorough safety assessments. The

absence of rigorous safety testing protocols and the reduction in human oversight demonstrates a disregard for the principle of beneficence, which requires proactive efforts to promote the well-being of society. There will be a huge lack of trust in AV technology if all companies prioritize progress over the safety of their customers.

Another ethical concern involves Uber's exploitation of Arizona's lenient regulatory environment. Arizona's minimal requirements for autonomous vehicle testing allowed Uber to deploy vehicles without stringent safety benchmarks. While Uber operated within the law, it failed to fulfill its ethical duty to exceed minimal regulatory standards when public safety was at stake. A deontological approach emphasizes adherence to higher moral principles, regardless of legal permissiveness. Uber's reliance on a permissive regulatory framework illustrates an ethical oversight, as corporations developing transformative technologies bear a heightened responsibility to safeguard the public, particularly when regulations lag behind innovation.

Although Uber's organizational and engineering failures are paramount, the role of the safety driver, Rafaela Vasquez, also warrants ethical scrutiny. Vasquez was found to have been streaming a television show on her smartphone at the time of the crash, violating her duty to remain vigilant. Safety drivers are ethically obligated to act as fail-safes for autonomous systems. However, Uber shares responsibility for this failure by not implementing adequate measures to monitor driver attentiveness, such as eye-tracking technology or periodic alerts. According to the Deontological framework, every entity, including the government, company, and employee, must follow the universal maxim that priotizes public safety.

Recommendations

To ensure safe operation, autonomous vehicles must integrate redundancies and fail-safes into their design. In the crash, Uber's vehicle relied solely on its proprietary system, disabling the

original manufacturer's emergency braking system. This lack of redundancy proved fatal. Future AV designs should mandate retaining safety features alongside proprietary systems, ensuring layered protections against critical failures. Additionally, advancements in sensor technology and perception algorithms should be prioritized to address blind spots in object recognition, particularly for detecting pedestrians in low-visibility conditions. Such improvements would help reduce false negatives, as occurred in this case when the system misclassified Elaine Herzberg as an "unknown object."

Safety protocols must become central to AV testing and deployment. Uber's decision to reduce the number of safety operators from two to one significantly increased risks, leaving the lone operator unsupervised when distracted. Companies should require at least two operators during testing phases, with clearly defined roles to monitor the system and the driving environment. Additionally, mandatory fatigue and distraction management protocols should be implemented for operators to minimize human error. Routine software and hardware audits are also crucial. These audits would evaluate system reliability and compliance with safety standards. Further, comprehensive pre-deployment simulations should stress-test AV systems in diverse and challenging scenarios, including edge cases, to ensure robust performance under real-world conditions.

Governments must establish and enforce comprehensive regulations for AV development and testing. Uniform national standards would provide consistency and accountability, reducing the patchwork of state-level regulations that currently exist in the United States. These standards should require companies to submit detailed safety assessments, including the results of rigorous testing protocols, before granting approval for public road testing. An independent oversight body should be established to monitor AV safety and investigate accidents. This body could

mandate transparency from AV developers, requiring them to share critical data and crash analyses to facilitate industry-wide learning. Regulators should also mandate adherence to safety-critical design practices, such as hazard analysis and risk assessment, as part of the certification process.

Companies must foster a safety-first culture, prioritizing ethical considerations over competitive pressures to deploy technologies rapidly. The 2018 crash highlighted Uber's neglect of safety in favor of aggressive timelines and cost-cutting measures. Leadership must embed safety and ethics into the company's mission, ensuring all stakeholders understand their responsibility to prioritize human lives over market advantages. Regular ethics training for engineers and operators should become standard practice, focusing on the moral implications of their work and the importance of adhering to safety protocols. Additionally, establishing independent ethics review boards within AV companies can provide oversight and accountability, particularly in high-stakes decision-making.

Collaboration among AV developers, regulators, and researchers is essential to advancing safety. Industry-wide forums can facilitate knowledge sharing on best practices, safety lessons, and emerging risks. For instance, sharing insights from accident investigations could help the industry collectively address common design flaws and vulnerabilities. Additionally, partnerships with academic institutions and government agencies can enhance research and development efforts, particularly in areas like sensor accuracy and ethical AI design. Public-private collaborations can also drive the creation of standardized safety benchmarks and testing protocols.

Conclusions

The 2018 Uber fatal crash marked a pivotal moment in the development and deployment of autonomous vehicle technology. The tragedy underscored critical shortcomings in technological systems, organizational priorities, and regulatory oversight, ultimately resulting in the preventable death of Elaine Herzberg. From a technological perspective, these failings highlight the need for layered redundancies, more robust perception algorithms, and rigorous testing in diverse conditions. Organizationally, a cultural shift within the industry is necessary, one that embeds safety and ethical responsibility as core principles. Regulatory bodies also bear responsibility for developing and enforcing a uniform, comprehensive regulatory framework for AV testing and deployment is critical. Ethically, the Uber fatal crash challenges society to rethink the balance between technological innovation and public safety. Developers, regulators, and operators must acknowledge their shared responsibility to protect human lives. Deontological framework emphasizes the intrinsic value of life and the moral obligation to avoid harm, providing valuable guidance for decision-making in AV design and deployment. By implementing technical improvements, strengthening safety protocols, enforcing robust regulations, and prioritizing ethics, the industry can mitigate risks and rebuild public trust. The lessons learned from this tragedy must guide the future of autonomous systems to prevent history from repeating itself.

Work Cited

Cummings, M. L., et al., 2021, Autonomous vehicle safety regulation: Progress, challenges, and recommendations: Journal of Transportation Safety and Security, v. 13, no. 1, pp. 1–20.

Goodman, J. and Halsey, A., 2018, Self-driving Uber car's fatal accident prompts scrutiny of testing standards: The Washington Post

Hawkins, A. J., 2018, Uber's self-driving car saw the pedestrian but didn't swerve – here's why: The Verge https://www.theverge.com/2018/5/24/17388696/uber-self-driving-crash-ntsb-report

Marshall, A., 2018, Uber's fatal self-driving crash: What really happened: Wired

McCausland, P., 2019, Self-Driving Uber Car That Hit and Killed Woman Did Not Recognize That Pedestrians Jaywalk. NBCNews.Com, NBCUniversal News Group, 9 Nov., www.nbcnews.com/tech/tech-news/self-driving-uber-car-hit-killed-woman-did-not-recognize-n1079281.

National Transportation Safety Board (NTSB), 2019, Collision between vehicle controlled by developmental automated driving system and pedestrian, Tempe, Arizona, March 18, 2018: NTSB Accident Report HWY18MH010, https://www.ntsb.gov/investigations/accidentreports/reports/har1903.pdf

Neuman, S., 2018, Arizona Governor Helped Make State 'wild West' for Driverless Cars." NPR, NPR, www.npr.org/sections/thetwo-way/2018/03/20/595115055/arizona-governor-helped-make-state-wild-west-for-driverless-cars.